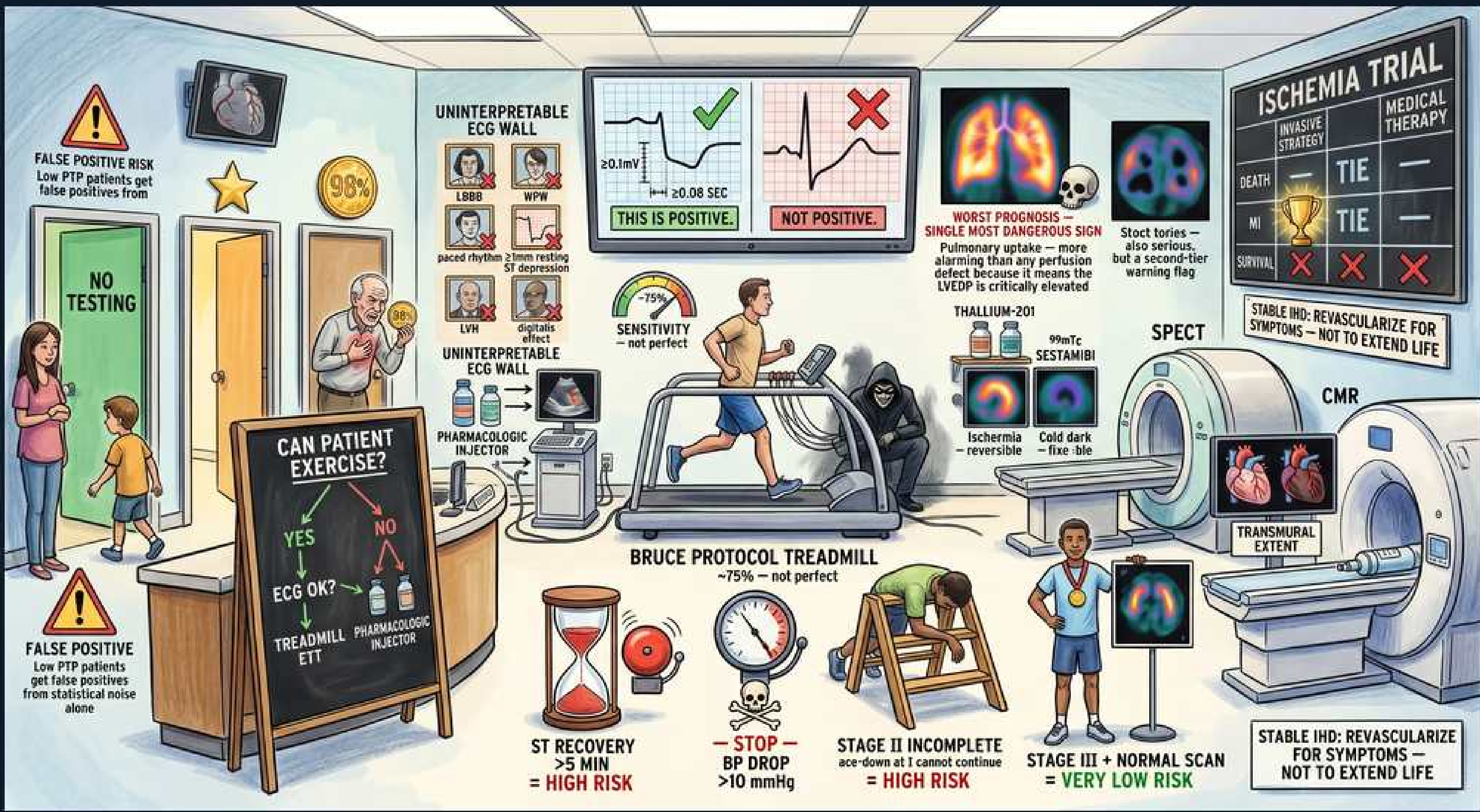
Stress Testing: Interpretation, High-Risk Markers & Pitfalls



1. False positives in low PTP

WHAT YOU SEE

FALSE POSITIVE RISK, low pre-test probability

WHAT IT ENCODES

In low pre-test-probability patients, most positive stress results are FALSE positives (Bayes: low prior → poor positive predictive value). This is why low-risk patients should not be stressed — a positive test misleads more than it informs.

BOARD TRAP

Acting on a positive stress test in a low-PTP patient — by Bayes most are false positives; PTP, not the result alone, drives action.

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2. No testing if PTP very low

WHAT YOU SEE

NO TESTING door

WHAT IT ENCODES

Very low pre-test probability patients need no testing — testing cannot meaningfully raise post-test probability and generates false positives leading to unnecessary angiography. Defer testing and reassess clinically.

BOARD TRAP

Reflex stress testing atypical low-risk chest pain — no testing is the correct answer when PTP is very low.

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3. ECG positive ≥1mm flat/downsloping

WHAT YOU SEE

ECG wall, ≥0.1mV / ≥0.08sec, 'THIS IS POSITIVE' vs 'NOT'

WHAT IT ENCODES

Exercise-ECG positivity = ≥1mm (0.1mV) horizontal or downsloping ST depression measured 0.08s after the J-point. Upsloping ST depression is NOT positive. This objective criterion separates true ischemic change from artifact.

BOARD TRAP

Calling upsloping ST depression ischemia — only flat/downsloping ≥1mm at 80ms past J-point qualifies.

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4. Uninterpretable ECG (LBBB/WPW/paced/LVH/digoxin)

WHAT YOU SEE

UNINTERPRETABLE ECG WALL: LBBB, WPW, paced, LVH, digoxin, ST depression

WHAT IT ENCODES

Baseline LBBB, ventricular pacing, WPW, LVH with strain, digoxin effect, or resting ST depression make the exercise ECG uninterpretable for ischemia. These patients need an imaging-based stress test (nuclear or echo) instead.

BOARD TRAP

Exercise ECG in LBBB/paced rhythm — the baseline obscures ST changes; use imaging stress (and vasodilator if LBBB).

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5. Sensitivity ~75% — not perfect

WHAT YOU SEE

Sensitivity gauge ~75%

WHAT IT ENCODES

Exercise ECG sensitivity is only ~75%, so a normal test does not exclude CAD, especially single-vessel or circumflex disease. Sensitivity improves with imaging (nuclear/echo). Use clinical judgment when suspicion is high.

BOARD TRAP

Excluding CAD after a negative exercise ECG — ~25% false-negative rate; high suspicion warrants imaging or angiography.

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6. Pulmonary uptake — worst prognosis

WHAT YOU SEE

Lung perfusion 'WORST PROGNOSIS, single most dangerous sign'

WHAT IT ENCODES

Increased lung (pulmonary) thallium uptake on perfusion imaging reflects stress-induced LV dysfunction and elevated filling pressures — it is the single most ominous prognostic marker, signaling extensive ischemia and high event risk.

BOARD TRAP

Overlooking lung uptake while focusing on defect size — pulmonary uptake is the strongest high-risk prognostic flag.

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7. Thallium-201 vs sestamibi

WHAT YOU SEE

Thallium-201 'reversible', Tc-99m sestamibi 'cold dark = fixed'

WHAT IT ENCODES

Thallium-201 redistributes, so a defect present on stress that fills in on rest = reversible ischemia; a persistent (fixed) cold defect = scar/infarct. Tc-99m sestamibi gives sharper images with less attenuation. Reversible = inducible ischemia.

BOARD TRAP

Labeling a fixed cold defect as active ischemia — fixed = scar; reversibility (redistribution) defines inducible ischemia.

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8. Bruce protocol treadmill ~75%

WHAT YOU SEE

Bruce Protocol treadmill runner

WHAT IT ENCODES

The Bruce protocol is the standard graded exercise treadmill test; achieving ≥85% of max predicted HR makes the study diagnostic. Exercise capacity (METs/Duke treadmill score) is itself prognostic — poor capacity predicts events.

BOARD TRAP

Reading a submaximal test (<85% MPHR) as truly negative — inadequate stress yields a non-diagnostic, falsely reassuring result.

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9. ST recovery >5 min = high risk

WHAT YOU SEE

Hourglass 'ST recovery >5 min = HIGH RISK'

WHAT IT ENCODES

ST-segment depression that persists more than 5 minutes into recovery indicates severe/extensive ischemia and is a high-risk finding warranting angiography. Prolonged ST recovery time correlates with worse anatomy and prognosis.

BOARD TRAP

Dismissing post-exercise ST changes — depression lasting >5 min into recovery is a high-risk marker, not benign.

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10. BP drop >10 mmHg = high risk

WHAT YOU SEE

Gauge 'STOP — BP drop >10 mmHg = HIGH RISK'

WHAT IT ENCODES

An exertional fall in systolic BP >10 mmHg signals exercise-induced LV dysfunction from severe multivessel or left main disease — stop the test; it is a high-risk indication for prompt angiography.

BOARD TRAP

Continuing a stress test despite a falling systolic BP — exertional hypotension means severe ischemia; terminate immediately.

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11. Stage III normal scan = very low risk

WHAT YOU SEE

Medal 'STAGE III + NORMAL SCAN = VERY LOW RISK'

WHAT IT ENCODES

Reaching Bruce stage III (high workload) with a normal perfusion scan and good exercise capacity indicates very low risk and excellent prognosis — no further testing or revascularization needed. Functional capacity is powerfully protective.

BOARD TRAP

Pursuing angiography after a high-workload normal scan — excellent exercise tolerance with normal imaging is very low risk.
